

MASHUK ELAHI

Electrical Engineer • Mission-Critical Power Systems & Critical Facilities

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PROFESSIONAL SUMMARY

Electrical engineer with 16+ years commissioning mission-critical LV/MV/HV power systems — FPSO topsides, offshore-wind substations, hybrid Li-ion BESS architectures and 300 mm semiconductor cleanrooms — applying a structured Level 1–6 commissioning framework (FAT review through performance/turnover) that maps directly onto Tier III/IV data-center and critical-facility delivery. Signature differentiators: IEC 61850 protection-relay/SCADA depth (ABB 800xA) and Python-automated EPMS load-list & cable-schedule generation at scale. Consistent pattern across every engagement — N+1/2N+1 redundancy, zero-tolerance commissioning, first-pass certification with zero rework.

CRITICAL-FACILITY COMMISSIONING FRAMEWORK (L1–L6)

L1 FAT Review → L2 Installation Oversight → L3 Energisation → L4 Functional Performance Testing (FPT) → L5 Integrated Systems Testing (IST) → L6 Performance / Turnover

SELECTED MISSION-CRITICAL CASE STUDIES

- ★ **MV 11kV Switchgear & ABB 800xA SCADA Fire & Gas Protection (L1–L5)** — Energised a heavy-industry MV/LV power system and SCADA-based fire & gas protection system with 100% MV/LV uptime through turnover and zero outstanding punch-list items, coordinating crews of 100+.
- ★ **IEC 61850 Protection-Relay Commissioning — Wind-Power Substation (L3–L5)** — Brought a wind-power substation to 100% IEC 61850 protection-relay conformity ahead of the energisation deadline; zero punch-list items at handover.
- ★ **300 mm Semiconductor Cleanroom Critical-Power Hook-Up & FPT (L3–L4)** — Energised critical power to IEC 60204/61511/ISO 14644 with zero audit rework and 100% tool-uptime/N+1 availability targets met; ~50% on-site installation labour cut via prefabricated power skids — directly transferable to critical-facility white-space fit-out.
- ★ **72 kWh BESS & EPMS Integration for a Hybrid Power Architecture (L2–L4)** — Integrated a Li-ion Battery Energy Storage System into an existing power architecture, cutting peak/low-load grid and fuel energy draw by up to 50% with zero rework at FAT.
- ★ **Python-Automated EPMS Load-List & Cable-Schedule Generation at Scale (automation)** — Designed and deployed Python scripts to auto-generate and cross-verify load lists and cable schedules against EPMS/point-to-point data, flagging discrepancies automatically — first-pass approval across four independent certification bodies with zero rework cycles; scales directly to large critical-facility EPMS point counts.
- ★ **Autonomous Remote Power Installation — Solar-Wind Hybrid Commissioning (L3–L5)** — Commissioned Malaysia's first autonomous, unmanned solar-wind remote power installation (24V DC solar/battery bus) with zero on-site operator presence — eliminated routine diesel use with zero unplanned shutdowns.

★ Strongest differentiators — full case-study portfolio: mnelahi-ai.github.io/portfolio/critical-facilities.html

CORE COMPETENCIES

- **Electrical Power Systems:** MV/HV/LV switchgear • power distribution • protection coordination • short-circuit, load-flow & arc-flash studies
- **Automation & Controls:** PLC • SCADA • IEC 61850 • EPMS/PMS • ABB 800xA • DP1/DP2
- **Hybrid & Critical Power:** Li-ion BESS • BMS • DC/DC inverters • N+1/2N+1 redundancy • shore power • DC/LVDC zonal distribution
- **Compliance & Commissioning:** FAT • SAT • FPT • IST • FMEA • cause & effect • IECEx/ATEX Zone 1/2 • ISO 9001/50001 • class approval (ABS/DNV/LR/BV)
- **Engineering Automation:** Python (EPMS, load-list & cable-schedule automation) • AI-assisted engineering workflows
- **Leadership:** Cross-functional teams of 100+ • mentoring • KPI tracking • continuous improvement

Core tools & software: ETAP • SKM PowerTools • MathCAD • MATLAB/Simulink • AutoCAD Electrical • ABB 800xA • Siemens/Yokogawa SCADA • Fluke DSX2-8000 • Python.

PROFESSIONAL EXPERIENCE

Electrical Engineer — Hybrid Marine Power

Jul 2024 – Present

Strategic Marine Pte Ltd • Singapore • Marine / Newbuild

- Architected the complete LV power system for a 42 m Gen-4 Fast Crew Boat (3 × CAT C32; 3 × 150 kW; VEEM propulsion) — SLDs, load lists, cable schedules — securing first-pass class approval across a programme portfolio spanning ABS, BV, LR & DNV.
- Integrated a 72 kWh Li-ion BESS, BMS, DC/DC inverters and a 90 kW shaft generator under LR Hybrid Power notation, cutting fuel burn by up to 50% in low-load and loitering modes; led BMS-to-PMS interface and FAT acceptance with the LR surveyor.
- Delivered full electrical design (SLDs, AMS I/O, cable schedules, IEC 60092-compliant IT earthing) across a multi-hull Gen-4 FCB programme, an offshore-wind CTV series, a 58 m Ro-Pax and a 43 m fast ferry — dual 380/220 V AC + 24 V DC, interlocked shore power to BV/LR/AMSA-NSCV.

- FAT-validated ABB azimuth-propulsion PMS on an autonomous-electric tug; led cross-fleet cathodic-protection and stray-current engineering; standardised departmental deliverables, KPI tracking and basis-of-design documentation.

Senior Project Engineer — Semiconductor & High-Tech Facilities

Feb 2023 – Jul 2024

Exyte Pte Ltd · Singapore · Semiconductor Cleanrooms

- Led 300 mm semiconductor cleanroom power hook-up to IEC 60204/61511/ISO 14644 — zero audit rework against 100% tool-up-time targets; PLC/SCADA interlock and functional-safety validation cleared first pass.
- Introduced prefabricated power skids that cut on-site labour by ~50% and eliminated contamination risk in a live cleanroom environment.
- Drove ISO 50001 energy stewardship through smart load monitoring, measurably reducing fab energy intensity; standardised site documentation as certified Lead Auditor.

E&I Engineer — Hazardous-Area, Renewables & Power

Jan 2021 – Jan 2023

Phoenix Mecano (S.E.A.) Pte Ltd · Singapore · Industrial / Offshore

- Pioneered Malaysia's first solar-wind autonomous unmanned wellhead platform (24 V DC solar/battery bus) — eliminated routine diesel use and enabled full remote operation, with zero unplanned shutdowns at commissioning.
- Directed an onshore cogeneration plant E&I scope — transformers, MV/LV switchgear, 400 kVA UPS, SLDs, tie-in and instrument-loop design; ABB/Siemens/Yokogawa SCADA integration and technical bid evaluation for all major E&I packages.
- Designed IECEx/ATEX Zone 1/2 hazardous-area systems — Ex equipment selection, IS barriers, zoning studies — and standardised compliance documentation as certified Lead Auditor (ISO 9001 & ISO 50001).

E&I Superintendent — FPSO, Rigs & Offshore Wind

Nov 2012 – Dec 2020

Keppel FELS (Keppel Offshore & Marine) · Singapore · Offshore / Marine

- Superintended crews of up to 100+ across fabrication, installation and integrated commissioning of FPSO topsides (Pre-FEED through detailed engineering and handover) — energised the disconnectable turret, MV 11 kV HV switchgear and ABB 800xA Fire & Gas; FPSO MV/LV reliability maintained at 100% uptime through sail-away.
- Energised an offshore-wind substation topside to 100% IEC 61850 conformity — full protection-relay coordination, alarm validation and sail-away sign-off; zero punch-list items outstanding at class handover.
- Commissioned ultra-deepwater jack-up drilling rigs and semisubmersible accommodation modules through FMEA, cause-and-effect matrices, FAT/SAT and sea trials; mentored 15+ engineers via Innovation & Quality Circles.
- Authored 30+ E&I risk assessments; delivered integrated bridge & navigation systems (gyro, VDR, alarm & watch); led EPC detailed-engineering deliverables for multiple concurrent vessel builds.

Assistant Electrical Engineer — Industrial & Infrastructure Projects

Sep 2009 – Oct 2012

Ananda Shipyard & Slipways Ltd · Bangladesh · Shipbuilding

- Led E&I crews of 15–30 on newbuild vessels and rigs — commissioned azimuth thrusters, controllable-pitch propellers and bridge navigation equipment to ABS/GL class and owner acceptance, achieving first-time milestone sign-off.
- Validated cable routing and interconnection designs against load schedules and voltage-drop criteria before energisation; governed BOM and delivery scheduling to protect on-site readiness against supply-chain disruption.
- Delivered LV/MV power distribution, protection coordination and commissioning across industrial and infrastructure projects — establishing the foundation of a 16-year, uninterrupted career in electrical engineering.

EDUCATION

- **MSc Sustainable Electricity Generation** — Universidad Católica de Murcia (UCAM), Spain · 2026
- **MSc Digital Transformation** — London Metropolitan University, UK · 2026
- **Master of Business Administration (MBA)** — University of Sunderland, UK · 2025
- **BSc Electrical & Electronic Engineering** — Bangladesh University · 2008
- **Diploma in Electronics Engineering** — Jessore Polytechnic Institute · 2004

CERTIFICATIONS & TRAINING

- IECEx Ex 001 — Protection in Explosive Atmospheres (valid to Oct 2027).
- Certified Lead Auditor (CQI IRCA) — ISO 9001:2015 Quality Management & ISO 50001:2018 Energy Management.
- Shipyard Supervisors Safety Course · WSQ WSH Management in Construction.
- Roxtec System Installer · CMP Cable Gland Installer · RAYCHEM APKT/EPKT.
- ABB KFELS B-Class Jack-up Rig LV Switchboard Training · Fluke DSX2-8000 Copper Certifier.
- Work at Height (Supervisor) · Confined Space Safety · WSH in Process Plant.